

```
# Install (run once)
!pip install torch transformers datasets peft accelerate
```

```
# Import libraries
import torch
from transformers import AutoModelForSequenceClassification, AutoTokenizer
from datasets import load_dataset
from peft import LoraConfig, get_peft_model
```

```
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: peft in /usr/local/lib/python3.12/dist-packages (0.18.1)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.13.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.7.1)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from peft) (5.9.5)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (3.13.3)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.4.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.24.1)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2026.2.25)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch) (3.0.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->dataset)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->dataset)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->dataset)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->dataset)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->dataset)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->dataset)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->dataset)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.12.1)
```

```
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (8.3.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (

```

```
model_name = "distilbert-base-uncased"
```

```
tokenizer = AutoTokenizer.from_pretrained(model_name)
```

```
model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
```

```
The secret `HF_TOKEN` does not exist in your Colab secrets.
```

```
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and resta
You will be able to reuse this secret in all of your notebooks.
```

```
Please note that authentication is recommended but still optional to access public models or datasets.
```

```
warnings.warn(
```

```
config.json: 100% 483/483 [00:00<00:00, 12.0kB/s]
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
```

```
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downl
```

```
tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 739B/s]
```

```
vocab.txt: 232k/? [00:00<00:00, 3.11MB/s]
```

```
tokenizer.json: 466k/? [00:00<00:00, 6.18MB/s]
```

```
model.safetensors: 100% 268M/268M [00:06<00:00, 69.8MB/s]
```

```
Loading weights: 100% 100/100 [00:00<00:00, 119.94it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
```

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
```

Key	Status
-----+-----	-----+-----
vocab_layer_norm.weight	UNEXPECTED
vocab_projector.bias	UNEXPECTED
vocab_layer_norm.bias	UNEXPECTED
vocab_transform.weight	UNEXPECTED
vocab_transform.bias	UNEXPECTED
classifier.bias	MISSING
pre_classifier.weight	MISSING
pre_classifier.bias	MISSING
classifier.weight	MISSING

```
Notes:
```

```
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
```

```
dataset = load_dataset("csv", data_files="/content/sentiment_dataset.csv")
```

```
def tokenize_function(example):
```

```
    return tokenizer(example["text"], truncation=True, padding="max_length")
```

```
tokenized_dataset = dataset.map(tokenize_function, batched=True)

train_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(tokenized_dataset["train"].num_rows))
test_dataset = tokenized_dataset["train"].shuffle(seed=42).select(range(tokenized_dataset["train"].num_rows))
```

Map: 100% 40/40 [00:00<00:00, 760.80 examples/s]

```
lora_config = LoraConfig(
    r=8,
    lora_alpha=16,
    target_modules=["q_lin", "v_lin"],
    lora_dropout=0.1,
    bias="none",
    task_type="SEQ_CLS"
)
```

```
peft_model = get_peft_model(model, lora_config)
```

```
peft_model.print_trainable_parameters()
```

WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernels will not work
trainable params: 739,586 || all params: 67,694,596 || trainable%: 1.0925

```
from transformers import TrainingArguments, Trainer
```

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
)
```

```
trainer = Trainer(
    model=peft_model,
    args=training_args,
    train_dataset=train_dataset
)
```

```
trainer.train()
```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but no accelerator is found, then device pin_memory will be ignored.
super().__init__(loader)

[5/5 01:25, Epoch 1/1]

Step Training Loss

```
TrainOutput(global_step=5, training_loss=0.693804121017456, metrics={'train_runtime': 125.4387, 'train_samples_per_second': 0.319, 'train_steps_per_second': 0.04,
'total_flos': 5389576273920.0, 'train_loss': 0.693804121017456, 'epoch': 1.0})
```

```
trainer.evaluate(eval_dataset=test_dataset)

text = "The movie was fantastic and inspiring!"
inputs = tokenizer(text, return_tensors="pt")

outputs = peft_model(**inputs)
prediction = torch.argmax(outputs.logits, dim=1)

print("Predicted label:", prediction.item())
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but no accelerator is found, then device pin_memory is set to false
  super().__init__(loader)

Predicted label: 1
```

Start coding or [generate](#) with AI.